

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		award (1) for percentage difference between the two values $\Rightarrow$ either 4.3% or 4.5% (1) students' decision is not valid award (1) for sensible explanation e.g. - total of percentage errors for both methods is 2.2% (1.4% + 0.8%) and the percentage difference is greater than this - total of percentage errors in titration and gravimetric analysis allow a difference of only 0.16×10^{-3} - values are $7.60 \pm 0.10 \times 10^{-3}$ and $7.27 \pm 0.06 \times 10^{-3}$ and these ranges do not overlap			2	2	2	
	(c)	(i)		royal blue / dark blue do not accept blue / pale blue or any reference to a precipitate	1			1		1
		(ii)		Indicative content - A negative ΔG means products are more likely to be present than reactants - Higher product concentration leads to large value of K_c - Formation of the Cu-cyclen complex breaks four Cu-ammonia bonds and forms four Cu-cyclen bonds - As the bonds broken and bonds made are of similar strengths enthalpy change will be small - For a negative ΔG, ΔS must be positive OR $T\Delta S$ must be greater than ΔH as $\Delta G = \Delta H - T\Delta S$ - The forward reaction produces more (aqueous) molecules as four NH_3 are released - The entropy of free ammonia would be higher than cyclen so ΔS would be positive		2	4	6	1	